DDTA research work plan after documentation-layer closure

WORK PLAN - REVISION 16

Status: active plan after BA3 integrated closure.

Supersedes: Revision 15 only for forward execution state; R1-R15 remain historical research records.

1. Fixed prior state

- Chapters 2-4: CLOSED / FINAL for current thesis scope.
- Documentation layer: CLOSED.
- BA0-R systems-modeling prior-art research: CLOSED.
- BA0 responsibility/non-goals: CLOSED.
- BA1 minimal BAE identity ontology: CLOSED.
- BA2 relation/action vocabulary: CLOSED BY BA2-T4.
- BA3 provenance/derivation/identity/lifecycle/change-revalidation mechanics: CLOSED BY BA3-T4.
- `BAReferent` and `BAProposition`: ACCEPTED first-class semantic identity families.
- ThreatForge is a case study/tooling instrument, not DDTA semantic authority.

2. BA3 closure result

BA3 closes the smallest current-scope contract for:

- independent baseline-scoped source provenance on both BA1 identity families;
- `GROUNDED | DERIVED | DIAGNOSTIC_UNRESOLVED` origin semantics;
- role-bound derivation basis and immutable inspectable rule revisions;
- `PENDING_REVIEW | ACCEPTED | REJECTED` review state;
- `CURRENT | STALE` freshness;
- family-specific cross-baseline semantic identity;
- `RETAIN | REPLACE | RETIRE` continuity with historical `SUPERSEDED/RETIRED` interpretations;
- explicit narrow `revalidationContext` and localized staleness propagation; and
- the governed-document authority gate through corrective feedback.

No third BAE family, new BA2 operator, universal derivation DSL, general dependency graph or implementation schema is forced.

The active closure artifact is:

- `methodology/BA3_PROVENANCE_DERIVATION_LIFECYCLE_CHANGE_CONTRACT_R1.md`

The BA3-T1/T2/T3 artifacts remain immutable research derivation history.

3. Base Analysis status after BA3

```
BA0   CLOSED
BA1   CLOSED
BA2   CLOSED
BA3   CLOSED
BA4   NOT STARTED / NEXT
BA5   NOT STARTED
BA6   NOT STARTED
```

The Base Analysis milestone as a whole is not yet closed. BA4 must establish projection semantics, BA5 lexical/assistance boundaries and BA6 integrated regression/holdout closure.

4. BA4 objective - projections

BA4 must define the smallest contract by which one accepted Base Analysis can support multiple derived human and methodology-oriented views without making any view a new source of project truth or allowing a consumer to redefine the shared core.

BA4 is not a system-modeling notation design and is not a formal STRIDE schema exercise.

Key questions include:

- what a projection may select, rename, group or aggregate from accepted BA semantics;
- what projection-local identity/traceability is needed back to BA elements;
- how a method-owned projection may add method-specific interpretation without feeding that taxonomy back into BA;
- which information must remain directly recoverable from BA so a consumer does not reconstruct project meaning from raw prose;
- how projection rebuild/revalidation follows BA change without becoming another lifecycle authority.

5. BA4-T1 - projection boundary, traceability and semantic-preservation lower-bound trial

NEXT / NOT EXECUTED BY THIS PLAN REVISION.

BA4-T1 must perform only the first projection-boundary trial.

It must pressure-test at least:

1. one human-oriented projection and one bounded method-oriented projection derived from the same accepted BA materialization;
2. selection/renaming/grouping/aggregation that preserves BA meaning versus projection-owned interpretation that must remain downstream;
3. the minimum trace from a projection item/result back to one or more `BAReferent`/`BAProposition` identities;
4. whether projection items require any new first-class BAE identity family or can remain view/method-owned constructs;
5. whether a consumer can obtain the needed shared project meaning without re-reading governed prose or importing method-specific semantics into BA;
6. how a BA identity replacement/retirement affects a projection without creating a second project lifecycle model;
7. whether any counterexample forces the smallest BA1/BA2/BA3 reopen.

BA4-T1 guardrails

Do not:

- define a complete STRIDE or STRIDE-AI ontology/schema;
- define AnalysisRecord or Common Finding;
- design ThreatForge classes/tables/UI;
- make a projection project authority;
- make DFD elements first-class BA types merely because one method consumes them;
- reopen Chapters 2-4 or BA0-BA3 without a material counterexample.

BA4-T1 exit condition

Produce a falsifiable projection responsibility/traceability lower-bound candidate and the next smallest pressure target. Do not close BA4 from one projection example.

6. BA5 - lexical vocabulary and optional assistance

Only after BA4 boundaries are stable, define display labels, authoring synonyms and optional assistance. Preserve:

```
source lexical wording
        !=
stable BA semantic key
        !=
display label / authoring synonym
```

NLP/LLM support remains candidate-producing assistance subject to provenance/review; it does not become semantic authority.

7. BA6 - complete Base Analysis regression and closure

Regress the complete BA0-BA5 design against the closed corpora and at least one structurally different holdout. BA6 may reopen only the smallest earlier responsibility on a material counterexample.

Only BA6 may close the complete Base Analysis milestone for the current thesis scope.

8. Analysis envelope remains downstream

Only after Base Analysis closure:

- A1 - AnalysisRecord / execution envelope;
- A2 - common reviewed Finding boundary;
- A3 - change/provenance integration;
- formal methodology overlays and STRIDE/STRIDE-AI evaluations.

9. Next authorized microstep

Execute only:

> **BA4-T1 - projection boundary, traceability and semantic-preservation lower-bound trial.**

Do not start BA5, BA6 or downstream analysis schemas before BA4-T1 is completed and reviewed.